Integrality of the five-step product recurrence (MathOverflow 323963)

Abstract

We prove that the rational sequence defined by

$$a_{n+5}a_n = \prod_{i=1}^{4}(a_{n+i}+1), \qquad a_0 = a_1 = a_2 = a_3 = a_4 = 1,$$

consists entirely of integers. Odd primes are controlled by a valuation-atom induction. The prime 2 is treated by an alternating first integral and a change of variables which reveals a fourteen-step 2-adic return.

1 Statement and initial values

All terms are positive rational numbers, so the recurrence is well-defined. The first terms are

$$1, 1, 1, 1, 1, 16, 136, 9316, 43398586, 941718655098991, \ldots.$$

Theorem 1. *For every $n \geq 0$, one has $a_n \in \mathbb{Z}$.*

Put

$$b_n = a_n + 1.$$

Since $a_n \in \mathbb{Q}_{>0}$, it is enough to prove

$$v_p(a_n) \geq 0 \qquad \text{for every prime } p \text{ and every } n. \tag{1}$$

2 Odd primes

Fix an odd prime p. Define integer-valued valuation atoms t_n by

$$t_0 = t_1 = t_2 = t_3 = t_4 = 0, \qquad t_{n+5} = v_p(b_{n+5}) - t_n \quad (n \geq 0). \tag{2}$$

Thus

$$v_p(b_{n+5}) = t_{n+5} + t_n. \tag{3}$$

2.1 The valuation identity

Lemma 2. *For every $n \geq 0$,*

$$v_p(a_{n+5}) = t_{n+1} + t_{n+2} + t_{n+3} + t_{n+4}. \tag{4}$$

Proof. The first five cases follow from

$$\begin{aligned} a_5 &= 16, \\ a_6 &= 8b_5, \\ a_7 &= 4b_5b_6, \\ a_8 &= 2b_5b_6b_7, \\ a_9 &= b_5b_6b_7b_8. \end{aligned} \tag{5}$$

Here the powers of 2 have zero p-adic valuation.

Suppose (4) holds at n. Taking valuations in the recurrence at $n+5$ and using (3) gives

$$\begin{aligned} v_p(a_{n+10}) &= \sum_{j=6}^{9} v_p(b_{n+j}) - v_p(a_{n+5}) \\ &= \sum_{j=6}^{9} (t_{n+j} + t_{n+j-5}) - \sum_{j=1}^{4} t_{n+j} \\ &= \sum_{j=6}^{9} t_{n+j}. \end{aligned}$$

This is (4) with n replaced by $n+5$. The five initial cases therefore cover all residue classes modulo 5. ☐

2.2 Nonnegativity and confinement

We prove simultaneously

$$t_m \geq 0 \tag{6}$$

and the confinement assertion

$$t_m > 0 \quad \Longrightarrow \quad t_{m-4} = t_{m-3} = t_{m-2} = t_{m-1} = 0. \tag{7}$$

For the initial check, the relevant values are

m	b_m
5	17
6	137
7	$9317 = 7 \cdot 11^3$
8	$43398587 = 29 \cdot 163 \cdot 9181$
9	$941718655098992 = 2^4 \cdot 37 \cdot 1172683 \cdot 1356497.$

(8)

The five displayed integers are pairwise coprime (equivalently, their odd parts are pairwise coprime). Hence a prime occurring in one of them occurs in none of the preceding four. Since $t_m = v_p(b_m)$ for $5 \leq m \leq 9$, this proves both (6) and (7) through $m = 9$.

Assume inductively that they hold through $m = k+9$. Lemma 2 already implies

$$v_p(a_{k+10}) = t_{k+6} + t_{k+7} + t_{k+8} + t_{k+9} \geq 0,$$

and hence $v_p(b_{k+10}) = v_p(1 + a_{k+10}) \geq 0$ by the non-Archimedean inequality.

Set $T = t_{k+5}$. If $T = 0$, then (2) immediately gives $t_{k+10} = v_p(b_{k+10}) \geq 0$. Suppose therefore that $T > 0$. Confinement at $k+5$ gives

$$t_{k+1} = t_{k+2} = t_{k+3} = t_{k+4} = 0,$$

so (4) shows that $v_p(a_{k+5}) = 0$.

Expanding the product of four successive b's gives the exact identity

$$\begin{aligned} b_{k+10}a_{k+5} = b_{k+5} + a_{k+6} + a_{k+7}b_{k+6} \\ + a_{k+8}b_{k+6}b_{k+7} + a_{k+9}b_{k+6}b_{k+7}b_{k+8}. \end{aligned}$$

Every summand on the right has valuation at least T. Indeed, $v_p(b_{k+5}) = t_k + T \geq T$. Formula (4) puts the atom T into the valuation of each of $a_{k+6}, \ldots, a_{k+9}$; all additional b-factors have nonnegative valuation because their two atoms in (3) are nonnegative. Thus

$$v_p(b_{k+10}) \geq T$$

because a_{k+5} is a p-adic unit. By (2), $t_{k+10} \geq 0$.

Finally, if $t_{k+10} > 0$, then (3) and $t_{k+5} \geq 0$ give $v_p(b_{k+10}) > 0$. Therefore $a_{k+10} = b_{k+10} - 1$ is a p-adic unit. Equation (4) now yields

$$0 = v_p(a_{k+10}) = t_{k+6} + t_{k+7} + t_{k+8} + t_{k+9}.$$

All four terms are nonnegative, so they are all zero. This is the next instance of (7), and the simultaneous induction is complete.

Combining (4) and (6) proves

$$v_p(a_n) \geq 0 \qquad (n \geq 0)$$

for every odd prime p.

3 The prime 2

The five initial numbers $b_0, \ldots, b_4$ are all even, so the preceding atom construction does not give nonnegative atoms at $p = 2$. We instead first reduce the recurrence by an exact alternating invariant.

3.1 The alternating invariant

Define

$$R_n = \frac{a_n a_{n+2} a_{n+4}}{a_{n+1}a_{n+3}(a_{n+1}+1)(a_{n+3}+1)}. \tag{10}$$

Lemma 3. *One has*

$$R_n = \begin{cases} \frac{1}{4}, & n \text{ even}, \\ 4, & n \text{ odd}. \end{cases} \tag{11}$$

Proof. Cancellation gives

$$R_n R_{n+1} = \frac{a_n a_{n+5}}{(a_{n+1}+1)(a_{n+2}+1)(a_{n+3}+1)(a_{n+4}+1)} = 1.$$

Moreover $R_0 = 1/4$, because $a_0 = \cdots = a_4 = 1$. Hence the values alternate as in (11). □

Write

$$\mathcal{T}(z) = \frac{z(z+1)}{2}, \qquad K_n = \begin{cases} 1, & n \text{ even}, \\ 16, & n \text{ odd}. \end{cases} \tag{12}$$

Since

$$a_{n+1}a_{n+3}(a_{n+1}+1)(a_{n+3}+1) = 4\mathcal{T}(a_{n+1})\mathcal{T}(a_{n+3}),$$

Lemma 3 is equivalent to

$$a_n a_{n+2} a_{n+4} = K_n \mathcal{T}(a_{n+1})\mathcal{T}(a_{n+3}). \tag{13}$$

3.2 The transformed recurrence

Define, for $n \geq 0$,

$$x_n = \frac{a_n a_{n+2}}{\mathcal{T}(a_{n+1})}. \tag{14}$$

All quantities are positive and nonzero. Multiplying (14) at n and $n+2$, and then using (13), gives

$$x_n x_{n+2} = K_n a_{n+2}. \tag{15}$$

In particular,

$$a_{n+2} = \frac{x_n x_{n+2}}{K_n}. \tag{16}$$

Eliminating the a's from (14)–(15) gives an autonomous recurrence with alternating coefficients:

$$x_{n+3} = \begin{cases} \dfrac{128 x_n x_{n+2}(1 + x_n x_{n+2})}{x_{n-1} x_{n+1}}, & n \text{ even}, \\ \dfrac{x_n x_{n+2}(16 + x_n x_{n+2})}{512 x_{n-1} x_{n+1}}, & n \text{ odd}. \end{cases} \tag{17}$$

For completeness, (17) follows by substituting $a_j = x_{j-2} x_j / K_j$ into the definition of x_{n+1} and cancelling x_{n+1}. The initial values are

$$x_0 = x_1 = x_2 = 1, \qquad x_3 = 16. \tag{18}$$

3.3 The fourteen-step return

Let

$$\mathcal{O} = \mathbb{Z}_2, \qquad \mathcal{U} = \mathbb{Z}_2^\times.$$

We use $2^r\mathcal{O}$ and $2^r\mathcal{U}$ also for negative integers r.

Lemma 4 (Fourteen-step 2-adic return)**.** *Let $x_0, x_1, x_2, x_3 \in \mathbb{Q}_2^\times$ satisfy*

$$x_0, x_1, x_2 \in \mathcal{U}, \qquad x_3 \in 16\mathcal{U}, \qquad x_0 \equiv x_2 \pmod 4. \tag{19}$$

Generate $x_4, \ldots, x_{17}$ by (17). Then

$$\begin{array}{c|cccccccccccccc} r & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 \\ \hline x_r \in & \mathcal{U} & \mathcal{U} & \mathcal{U} & 2^4\mathcal{U} & \mathcal{O} & 2^4\mathcal{O} & 2^2\mathcal{O} & 2^2\mathcal{U} & 2^{-2}\mathcal{U} & 2^2\mathcal{U} & 2^2\mathcal{O} & 2^4\mathcal{O} & \mathcal{O} & 2^4\mathcal{U}. \end{array} \tag{20}$$

At the end of the block,

$$x_{14}, x_{15}, x_{16} \in \mathcal{U}, \qquad x_{17} \in 16\mathcal{U}, \qquad x_{14} \equiv x_{16} \equiv -x_0 \pmod 4. \tag{21}$$

In particular, the hypotheses (19) return after fourteen steps.

Proof. We give the complete finite calculation. Write

$$x_0 = u_0, \qquad x_1 = u_1, \qquad x_2 = u_2, \qquad x_3 = 16u_3, \tag{22}$$

where all u_i are units and $u_2 - u_0 \in 4\mathcal{O}$. Normalize the remaining terms according to (20)–(21):

$$\begin{array}{c|cccccccccccccc} r & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 & 17 \\ \hline x_r & z_4 & 16z_5 & 4z_6 & 4z_7 & z_8/4 & 4z_9 & 4z_{10} & 16z_{11} & z_{12} & 16z_{13} & z_{14} & z_{15} & z_{16} & 16z_{17}. \end{array} \tag{23}$$

Substitution in (17), with cancellation at each preceding line, gives the following straight-line identities:

$$\begin{aligned}
z_4 &= \frac{u_1u_3(1+u_1u_3)}{2u_0u_2},\\
z_5 &= \frac{u_2z_4(1+u_2z_4)}{2u_1u_3},\\
z_6 &= \frac{(1+u_2z_4)(1+16u_3z_5)}{u_1},\\
z_7 &= \frac{(1+16u_3z_5)(1+4z_4z_6)}{u_2},\\
z_8 &= \frac{(1+4z_4z_6)(1+4z_5z_7)}{u_3},\\
z_9 &= \frac{z_6z_8(1+z_6z_8)}{2z_5z_7},\\
z_{10} &= \frac{z_7z_9(1+z_7z_9)}{8z_6z_8},\\
z_{11} &= \frac{z_8z_{10}(1+z_8z_{10})}{2z_7z_9},\\
z_{12} &= \frac{2z_9z_{11}(1+4z_9z_{11})}{z_8z_{10}},\\
z_{13} &= \frac{z_{10}z_{12}(1+4z_{10}z_{12})}{2z_9z_{11}},\\
z_{14} &= \frac{2z_{11}z_{13}(1+16z_{11}z_{13})}{z_{10}z_{12}},\\
z_{15} &= \frac{z_{12}z_{14}(1+z_{12}z_{14})}{2z_{11}z_{13}},\\
z_{16} &= \frac{z_{13}z_{15}(1+z_{13}z_{15})}{2z_{12}z_{14}},\\
z_{17} &= \frac{z_{14}z_{16}(1+z_{14}z_{16})}{2z_{13}z_{15}}.
\end{aligned}$$

Although some denominators in the second half of (24) need not be units, all cancellations below are exact. We now give the finite certificate in the form used in the formal proof.

Put

$$\alpha = v_2(z_4), \qquad \gamma = v_2(1+u_2z_4),$$

and define the three exceptional quotients

$$q_{10} = \frac{1+z_7z_9}{8z_6}, \qquad q_{15} = \frac{1+z_{12}z_{14}}{z_{10}}, \qquad q_{16} = \frac{1+z_{13}z_{15}}{2z_{12}}. \tag{25}$$

We shall prove

$$\begin{aligned}
&z_4, z_5, z_6 \in \mathcal{O}, \qquad z_7, z_8, z_9 \in \mathcal{U},\\
&q_{10} \in \mathcal{O}, \qquad q_{15}, q_{16} \in \mathcal{U},\\
&z_{13}, z_{14}, z_{15}, z_{16}, z_{17} \in \mathcal{U},\\
&z_{14} \equiv z_{16} \equiv -u_0 \pmod 4.
\end{aligned}$$

First, a unit squared is congruent to 1 modulo 8, and the sum of two 2-adic units has positive valuation. The first two lines of (24) therefore give

$$v_2(z_5) = \alpha + \gamma - 1, \qquad v_2(z_6) = \gamma.$$

Here $\alpha \geq 0$. If $\alpha = 0$, then $\gamma \geq 1$; if $\alpha > 0$, then $\gamma = 0$ and $\alpha \geq 1$. Hence $z_4, z_5, z_6 \in \mathcal{O}$.

Set

$$A = 1 + u_2 z_4, \quad B = 1 + 16 u_3 z_5,$$
$$C = 1 + 4 z_4 z_6, \quad D = 1 + 4 z_5 z_7.$$

The element B is a unit. Since $z_4, z_6 \in \mathcal{O}$, the element $C = 1 + 4z_4z_6$ is a unit as well. The relevant identities are

$$z_6 = \frac{AB}{u_1}, \qquad z_7 = \frac{BC}{u_2}, \qquad z_8 = \frac{CD}{u_3}.$$

It follows successively that z_7 is a unit, then that $D = 1 + 4z_5z_7$ is a unit, and finally that z_8 is a unit. Moreover

$$E := \frac{u_1u_3 + ABCD}{z_4} \in \mathcal{U}, \tag{27}$$

and the sixth line of (24) reduces exactly to

$$z_9 = \frac{DE}{u_1u_3} \in \mathcal{U}. \tag{28}$$

To verify (27), if z_4 is a unit then A is even, so the numerator in (27) is a unit; division by the unit z_4 preserves this. If $t = v_2(z_4) > 0$, then

$$BCD \equiv 1 \pmod{2^{t+1}}, \qquad A = 1 + u_2 z_4.$$

Writing $p_0 = u_1u_3$, the first line of (24) gives

$$p_0 + 1 = \frac{2u_0u_2z_4}{p_0}.$$

Consequently

$$p_0 + ABCD \equiv u_2 z_4 \left(1 + \frac{2u_0}{p_0}\right) \pmod{2^{t+1}},$$

whose coefficient of z_4 is a unit. This proves (27)–(28).

We next prove that q_{10} is integral. Put

$$s = \frac{2z_4A}{p_0}, \qquad Q = BCD, \qquad L = \frac{Q-1}{s}.$$

The preceding straight-line identities give

$$B = 1 + 4u_2u_3s, \qquad C = 1 + 2u_3sB, \qquad D = 1 + sBC,$$

and hence, exactly,

$$L = 4u_2u_3 + 2u_3B^2 + (BC)^2.$$

Thus $L \equiv -1 \pmod 4$. Also

$$p_0 \equiv 2z_4 - 1 \pmod 4, \qquad s \equiv 2z_4A \pmod 4.$$

A direct factorization of the numerator of $1 + z_7 z_9$ now gives

$$q_{10} = \frac{F\,u_1}{4u_2 p_0^2 B}, \qquad F = u_0 u_2 + (p_0 + 2A)L + AsL^2.$$

All factors outside $F/4$ are units. Using $u_0 \equiv u_2 \pmod 4$ and the preceding congruences,

$$F \equiv 2z_4(A^2 - u_2 - 1) \pmod 4.$$

Since

$$A^2 - u_2 - 1 = u_2(2z_4 + u_2 z_4^2 - 1),$$

the right side is divisible by 4: if $\alpha > 0$, the factor $2z_4$ already supplies two powers of 2; if $\alpha = 0$, then both terms inside the final parenthesis are even. Hence

$$v_2(F) \geq 2, \qquad q_{10} \in \mathcal{O}.$$

The same identities also give the returned residue of z_9:

$$z_9 = \frac{D}{p_0^2}\left(2u_0 u_2 + p_0 u_2 + 2A^2 L\right).$$

Reduction modulo 4, using $L \equiv -1$, gives

$$z_9 \equiv -u_2 \equiv -u_0 \pmod 4.$$

The only apparent correction is $2z_4(u_2 - z_4)$; it is divisible by 4, since either z_4 is even or both u_2 and z_4 are odd.

For the second half of the block, write

$$r = q_{10}, \qquad P = z_7 z_9,$$

and introduce

$$\begin{aligned} H &= 1 + Pr, \\ I &= 1 + 2z_9 rH, \\ J &= 1 + 4\frac{z_9}{z_8} rHI, \\ K &= 1 + 8\frac{1}{z_8} rHIJ. \end{aligned}$$

Since $r \in \mathcal{O}$, the element H is integral and I, J, K are units. Exact simplification of (24) yields

$$\begin{aligned} z_{10} &= \frac{Pr}{z_8}, \quad z_{11} = \frac{rH}{2}, \quad z_{12} = \frac{HI}{z_7}, \\ z_{13} &= \frac{IJ}{z_8}, \quad z_{14} = \frac{JK}{z_9}. \end{aligned}$$

Define

$$L_{15} = \frac{HIJK - 1}{r}.$$

Successive expansion of I, J, K gives

$$L_{15} = P + 2z_9 H^2 + 4\frac{z_9}{z_8}(HI)^2 + 8\frac{1}{z_8}(HIJ)^2.$$

The first summand is a unit and all the others have positive valuation, so L_{15} is a unit. The exact formula

$$q_{15} = \frac{z_8}{P^2}(8z_6 + L_{15}) = \frac{z_8}{P^2}\frac{P + HY}{r}.$$

therefore proves $q_{15} \in \mathcal{U}$. The next normalized term satisfies

$$z_{15} = Kq_{15} \in \mathcal{U}.$$

Moreover $J \equiv K \equiv 1 \pmod 4$, so

$$z_{14} = \frac{JK}{z_9} \equiv -u_0 \pmod 4.$$

It remains to prove that q_{16} is a unit. Put

$$Y = IJK, \qquad T = \frac{Y-1}{H}, \qquad W = P + Y^2 + PT.$$

Exact expansion gives

$$T = 2z_9 r + 4\frac{z_9}{z_8} r I^2 + 8\frac{1}{z_8} r (IJ)^2.$$

The first summand has valuation $v_2(r)+1$, while the other two have valuation at least $v_2(r)+2$ and $v_2(r)+3$. Thus

$$v_2(T) = v_2(r) + 1.$$

Since $Y - 1 = HT$, Y is a unit, and $Y + 1$ has positive valuation,

$$v_2(Y^2 - 1) \geq v_2(r) + 2.$$

On the other hand, $P + 1 = 8z_6 r$, so

$$v_2(P+1) \geq v_2(r) + 3.$$

Consequently $v_2(P + Y^2) \geq v_2(r) + 2$, whereas $v_2(PT) = v_2(r) + 1$. Therefore

$$v_2(W) = v_2(r) + 1.$$

Finally, exact cancellation in (24) gives

$$q_{16} = \frac{W z_7}{2IP^2 r}.$$

The numerator and denominator have the same valuation $v_2(r)+1$, proving $q_{16} \in \mathcal{U}$.

We still need the returned residue of z_{16}. Put

$$S = q_{15}, \qquad M = z_{13}z_{15} = \frac{YS}{z_8}.$$

Then

$$z_{16} = \frac{Mq_{16}}{z_{14}}, \qquad q_{16} = \frac{1+M}{2z_{12}}.$$

Because z_{14} is a unit and $z_{14}^2 \equiv 1 \pmod 8$, it is enough to prove

$$Mq_{16} \equiv 1 \pmod 4. \tag{29}$$

Suppose first that $v_2(r) > 0$. Then $P \equiv -1 \pmod{16}$. The exact formula for q_{15} gives

$$\frac{P + HY}{r} \equiv P + 2z_9 + 4\frac{z_9}{z_8} \pmod{8}.$$

This follows by substituting the displayed expansions of I, J, K; together with $P \equiv -1 \pmod{16}$, it gives $M \equiv 1 \pmod{4}$. If $v_2(r) \geq 2$, reduction of the exact formulas modulo 4 gives

$$\frac{M-1}{2} \equiv z_9 + 1 \equiv z_7 - 1, \qquad z_{12} \equiv z_7.$$

If $v_2(r) = 1$, the additional first-order contribution of I gives

$$\frac{M-1}{2} \equiv z_9 + 1 + z_9 rH, \qquad z_{12} \equiv z_7 + z_9 r.$$

In both subcases,

$$1 + \frac{M-1}{2} \equiv z_{12} \pmod{4},$$

and hence $M(1 + M)/2 \equiv z_{12} \pmod{4}$. This proves (29).

Now suppose that $v_2(r) = 0$. Then r is a unit and H has positive valuation. The exact formula for q_{15} gives

$$M = \frac{Y(P + HY)}{P^2 r} \equiv P \equiv -1 \pmod{4}.$$

Writing $T = (Y - 1)/H$ as above and setting

$$V = \frac{1}{2}(P + Y^2 + PT),$$

the expansion of T, together with $P + 1 \equiv 0 \pmod{8}$, gives

$$V \equiv P(z_9 r + 2) \pmod{4}.$$

Since

$$q_{16} = \frac{z_7 V}{P^2 r I},$$

and $P^2 \equiv I \equiv 1 \pmod{4}$, this reduces to

$$q_{16} \equiv \frac{z_7(z_9 r + 2)P}{r} \equiv -1 \pmod{4}.$$

Thus $Mq_{16} \equiv 1 \pmod{4}$, proving (29) in the second case as well. It follows that

$$z_{16} \equiv z_{14} \equiv -u_0 \pmod{4}.$$

For completeness, let

$$\delta = v_2(z_{10}), \qquad \epsilon = v_2(z_{11}), \qquad \zeta = v_2(z_{12}).$$

Because P, z_8 are units, the formula for z_{10} gives $\delta = v_2(r) \geq 0$. The next two formulas give

$$\epsilon = \delta + v_2(1 + z_8 z_{10}) - 1, \qquad \zeta = 1 + \epsilon - \delta.$$

If $\delta > 0$, then $v_2(1 + z_8 z_{10}) = 0$, so $\epsilon = \delta - 1 \geq 0$ and $\zeta = 0$. If $\delta = 0$, then $z_8 z_{10}$ is a unit, so $v_2(1 + z_8 z_{10}) \geq 1$; hence $\epsilon \geq 0$ and $\zeta = 1 + \epsilon \geq 1$.

The unit properties of q_{15} and q_{16} say precisely that

$$v_2(1 + z_{12} z_{14}) = \delta, \qquad v_2(1 + z_{13} z_{15}) = 1 + \zeta.$$

Thus $z_{13}, z_{14}, z_{15}, z_{16}$ are units. Finally, $z_{14} z_{16} \equiv 1 \pmod{4}$, so $v_2(1 + z_{14} z_{16}) = 1$; the last line of (24) then makes z_{17} a unit. This establishes all of (26), hence (20)–(21). $\square$

3.4 Completion of the 2-adic argument

The initial transformed window (18) satisfies the hypotheses of Lemma 4. Because the length 14 is even, the parity in (17) is unchanged when the lemma is iterated. Hence, for all $n \geq 0$,

$$v_2(x_n) \geq \ell_{n \bmod 14}, \tag{30}$$

where

$$(\ell_0, \dots, \ell_{13}) = (0, 0, 0, 4, 0, 4, 2, 2, -2, 2, 2, 4, 0, 4). \tag{31}$$

By (16),

$$v_2(a_{n+2}) = v_2(x_n) + v_2(x_{n+2}) - v_2(K_n). \tag{32}$$

For even n, $v_2(K_n) = 0$; for odd n, $v_2(K_n) = 4$. Substitution of (31) into the right side of (32) gives, as n runs through the residue classes modulo 14,

$$(0, 0, 0, 4, 2, 2, 0, 0, 0, 2, 2, 4, 0, 0). \tag{33}$$

Every entry is nonnegative. Therefore

$$v_2(a_n) \geq 0 \qquad (n \geq 0). \tag{34}$$

4 Conclusion

For every n, the positive rational number a_n has nonnegative valuation at every odd prime by Section 2 and at the prime 2 by (34). Its reduced denominator therefore has no prime divisor, so it is 1. This proves Theorem 1. Every identity and valuation argument used above, including the finite cancellation certificate (26), has also been checked in Lean 4 with Mathlib.

Remark 5. The single negative entry -2 in the transformed valuation word (31) is harmless. Formula (16) recovers a_{n+2} from a product of two transformed terms, and the paired lower bounds in (33) are all nonnegative. The exceptional high spikes seen in the sequence $v_2(a_n)$ merely increase some of the non-unit variables $z_4, z_5, z_6, z_{10}, z_{11}, z_{12}$; they do not disturb the fourteen-step return tube.